Slide 1 · Session opening

Main point: Understand what additional training changes before choosing a method.

We have spent the morning choosing model and workflow components and supplying information. We now ask a different question: can we change how the model responds? Post-training means starting with an existing model and training it further. We will separate the behavior we reward, the parameters we allow to change and the evidence needed to judge the result. These are three decisions, not a competition between acronyms. First we learn from reviewed answers. Then we make the update cheaper. Finally we learn from comparisons between answers.

Example to use: A document can be present in the prompt and still be misread. Retrieval solves access to that document; training might help a recurring interpretation problem.

Assumptions and limits: This is a conceptual introduction. No particular dataset or model improvement is assumed. The small examples illustrate mechanics and are not experimental results.

Ask the room: Which part of your current workflow is difficult because information is missing, and which part is difficult even when the information is present?

Slide 2 · What changes in Section 3?

Main point: Distinguish changing the input from changing the model.

Adding a document to a prompt changes what the model can read on this call. It does not ordinarily change its stored parameters. Further training changes those parameters and can influence later answers even when the training examples are no longer in the prompt. This is why the earlier retrieval work remains necessary. Training cannot establish the contents of a new document it has never received. Here we explain the training choices. In the next section we will inspect or run the process and compare its effects.

Example to use: Pasting a newly announced meeting date gives the model current context. Teaching many examples of identifying the latest date may change the skill it applies to later documents.

Assumptions and limits: We are discussing ordinary inference versus parameter updates. A product may separately store conversation memory; that is not automatically a weight update.

Ask the room: If the prompt is removed, which kind of change can remain in the parameters?

Slide 3 · Five things to be able to explain

Main point: Know the five practical questions this session will answer.

SFT stands for supervised fine-tuning: learning from demonstrations. LoRA stands for low-rank adaptation: representing a trainable change with relatively small matrices. QLoRA adds quantized storage of the frozen base. DPO means direct preference optimization: using a chosen and rejected answer together. Conventional RLHF uses human feedback to learn a reward and then optimize generated responses against it. Each term will be introduced through the problem it solves, before the equation. We finish by asking whether the behavior has improved on cases that were not used for training or tuning.

Example to use: You should be able to tell a colleague why a small adapter can be trained with either demonstrations or preferences without claiming that those are the same learning signal.

Assumptions and limits: These are explanatory objectives. Actual effectiveness depends on data, model, optimization and the evaluation design.

Ask the room: Which acronym is least familiar? Use that answer to decide where to slow down.

Slide 4 · Start with three separate questions

Main point: Keep training objective, parameter update and storage precision separate.

The first question is what a better answer looks like. A demonstration or preference label supplies that information. The second question is which numbers inside the model are allowed to change. We might change nearly all weights or add a small LoRA correction. The third question is how precisely the large frozen base is stored. Lower precision can save memory. These choices can be combined. Saying we use LoRA does not tell us whether we learn from ideal answers or preferred pairs, and saying we use SFT does not tell us whether every base weight will change.

Example to use: SFT plus LoRA means learning from demonstrated responses while updating adapter matrices. SFT plus QLoRA uses the same kind of learning signal with a quantized frozen base.

Assumptions and limits: Compatibility requires an implementation that supports the selected model and numerical configuration. These combinations are conceptual options, not guaranteed equivalent results.

Ask the room: Does the word LoRA tell us what the training label is?

Slide 5 · SFT: supervised fine-tuning - 1 of 3

Main point: Provide a reviewed demonstration

A record contains the task, input and response we want the model to learn. The target response is the label. A demonstration is an example of the response we want. The model is shown the input and reviewed answer. Training rewards placing more probability on the answer tokens in their context. The model is not given an abstract rule saying be correct and left to discover a verifier. It learns from the supplied targets. This can teach task structure, tone or a recurring interpretation pattern, but it can also reproduce errors and omissions in those targets. A reviewer therefore needs a clear standard for what makes a demonstration acceptable.

Example to use: Input: Extract the day from The meeting is on Tuesday. Reviewed answer: Tuesday. Another record can teach that a postponed meeting uses the replacement date.

Assumptions and limits: We assume targets represent the behavior we intend to imitate. That is a data-quality assumption, not a mathematical guarantee that each target is true.

Ask the room: If every reviewed response contains an unnecessary confident statement, what behavior are we rewarding?

Slide 6 · SFT: supervised fine-tuning - 2 of 3

Main point: Reward the recorded answer tokens

Training increases their conditional likelihood. It does not automatically check whether the reviewer answer is true. A demonstration is an example of the response we want. The model is shown the input and reviewed answer. Training rewards placing more probability on the answer tokens in their context. The model is not given an abstract rule saying be correct and left to discover a verifier. It learns from the supplied targets. This can teach task structure, tone or a recurring interpretation pattern, but it can also reproduce errors and omissions in those targets. A reviewer therefore needs a clear standard for what makes a demonstration acceptable.

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Ask the room: If every reviewed response contains an unnecessary confident statement, what behavior are we rewarding?

Slide 7 · SFT: supervised fine-tuning - 3 of 3

Main point: Test on different cases

A model can fit the demonstrations without handling new wording or completing the whole workflow correctly. A demonstration is an example of the response we want. The model is shown the input and reviewed answer. Training rewards placing more probability on the answer tokens in their context. The model is not given an abstract rule saying be correct and left to discover a verifier. It learns from the supplied targets. This can teach task structure, tone or a recurring interpretation pattern, but it can also reproduce errors and omissions in those targets. A reviewer therefore needs a clear standard for what makes a demonstration acceptable.

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Ask the room: If every reviewed response contains an unnecessary confident statement, what behavior are we rewarding?

Slide 8 · What one training record contains

Main point: Show exactly where the input and label live.

The instruction says what to do. The input supplies the text or other context. The target response is the label to imitate. The training library uses a chat template or prompt-completion format to serialize these parts into tokens. A label here can be an entire sentence or structured response; it does not have to be a numeric class. Before training, inspect several records in a human-readable form. Check that answers correspond to the correct inputs and that the template is consistent with the model.

Example to use: Instruction: Extract the meeting day. Input: The meeting is on Tuesday. Completion: Tuesday. The completion field is the supervised target.

Assumptions and limits: This is a prompt-completion illustration. Conversational datasets use role-marked messages, and the loss configuration must match the chosen format.

Ask the room: Which field would you correct if a reviewer accidentally entered Wednesday?

Slide 9 · Teacher forcing: predict each next target token

Main point: Explain what context is available during training.

The model predicts one token at a time. During training we supply the correct preceding answer tokens from the record. This is called teacher forcing. It lets us calculate a prediction error at every selected position. During normal generation, the model instead uses the tokens it has actually generated, including any earlier mistake. That difference is why low training loss is not the same as demonstrating successful free-running behavior. We need both a training signal and a separate test in which the model produces the answer itself.

Example to use: For an answer such as The meeting is Tuesday, the model predicts each next token after the correct earlier prefix during training. In a live answer its own prefix might already be wrong.

Assumptions and limits: The autoregressive model conditions on earlier tokens. It does not assume tokens in a response are independent. Actual token boundaries can split words and punctuation.

Ask the room: Could a model score well with correct prefixes but make mistakes after its own first wrong token?

Slide 10 · A small labelled SFT dataset

Main point: Make the label concrete with two different examples.

Read the input and target across each row. The first record teaches direct extraction. The second requires recognizing that the replacement date is the current one. A useful dataset includes the variations the intended task actually contains rather than hundreds of copies of the easiest form. The target is still the reviewed response, not a probability supplied by the model. Keep original source identifiers so related examples can be grouped when splitting training and evaluation data.

Example to use: Record one has input The meeting is on Tuesday and label Tuesday. Record two has input The meeting moved from Tuesday to Friday and label Friday.

Assumptions and limits: These two records are illustrative and are not enough to train or validate a robust model. Label correctness and coverage need review. Paraphrases from one source should not be treated as independent test families.

Ask the room: What additional record would test a cancellation or an unspecified date?

Slide 11 · The loss mask selects what is scored - 1 of 3

Main point: Keep the question visible

The model still reads the instruction and input while predicting the response. A loss mask is an inclusion switch. In the response-only setup, a target response position gets one and a prompt or padding position gets zero. Zero means that position has no direct term in the loss. It does not mean the model cannot read the instruction. The prompt still influences predictions of the scored response tokens, and training gradients can pass through the computations that use it. An attention mask answers a different question: which positions can interact. Keep those two meanings separate.

Example to use: The model reads Extract the day and The meeting is on Tuesday, but the direct prediction error is calculated on the completion Tuesday and its declared termination tokens.

Assumptions and limits: Response-only scoring is our chosen configuration, not a universal property of SFT. Some setups score other parts of the sequence.

Ask the room: If the prompt has zero direct loss, can changing it still change the response loss? Yes, because it changes the context.

Slide 12 · The loss mask selects what is scored - 2 of 3

Main point: Score the selected answer positions

A mask of one includes a prediction error. A mask of zero excludes that position from direct loss. A loss mask is an inclusion switch. In the response-only setup, a target response position gets one and a prompt or padding position gets zero. Zero means that position has no direct term in the loss. It does not mean the model cannot read the instruction. The prompt still influences predictions of the scored response tokens, and training gradients can pass through the computations that use it. An attention mask answers a different question: which positions can interact. Keep those two meanings separate.

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Ask the room: If the prompt has zero direct loss, can changing it still change the response loss? Yes, because it changes the context.

Slide 13 · The loss mask selects what is scored - 3 of 3

Main point: Do not confuse this with attention

A loss mask selects errors to score. An attention mask controls which positions can interact. A loss mask is an inclusion switch. In the response-only setup, a target response position gets one and a prompt or padding position gets zero. Zero means that position has no direct term in the loss. It does not mean the model cannot read the instruction. The prompt still influences predictions of the scored response tokens, and training gradients can pass through the computations that use it. An attention mask answers a different question: which positions can interact. Keep those two meanings separate.

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Ask the room: If the prompt has zero direct loss, can changing it still change the response loss? Yes, because it changes the context.

Slide 14 · Before the equation: what is assumed?

Main point: State the conventions that make the formula interpretable.

The formula will describe a conditional next-token model. It predicts each target using its preceding context. The reviewed response is treated as the target to imitate. We use teacher forcing, exclude prompt and padding positions from direct scoring and average over the remaining tokens. These are explicit choices. There are other reasonable ways to weight examples or score conversations. The loss can be computed even if records are correlated; stronger statistical claims about performance need an appropriate sampling and evaluation design.

Example to use: Two long demonstrations contribute more scored tokens than two short ones under the token-weighted average shown here. Equal-example weighting would be a different convention.

Assumptions and limits: Targets must be suitable demonstrations; tokenization and masks must align; the denominator counts scored tokens and must be positive. No independent-token assumption is made.

Ask the room: Would you want a long boilerplate answer to receive more weight simply because it has more tokens?

Slide 15 · SFT loss: one average over scored tokens

Main point: Read every part of the equation in ordinary language.

Theta names the current trainable parameters. The index i selects a training record and t selects a token position in that record. The model assigns a probability to the recorded token given everything before it. Taking the negative logarithm gives a larger penalty to low probability. Multiplying by the mask includes or excludes that term. The inner sum adds positions, the outer sum adds records, and M divides by the total number of scored positions. Minimizing this average encourages higher probability on the demonstrated answers.

Example to use: A target token assigned probability 0.8 receives a smaller penalty than the same target assigned probability 0.2. The following slide turns that into numbers.

Assumptions and limits: This is a token-weighted empirical loss with response-only masks and teacher forcing. It is not an event-probability calibration objective. Training-record dependence does not stop arithmetic, but affects generalization arguments and effective sample size.

Ask the room: Why is M the scored-token count rather than the total number of input tokens?

Slide 16 · A tiny calculation makes the loss concrete

Main point: Demonstrate what the average actually computes.

There are two scored target positions in this illustration. The first gets probability 0.8, whose negative natural logarithm is about 0.223. The second gets probability 0.5, giving about 0.693. Their average is 0.458. Improving either target probability lowers the average loss. The numbers concern predicting recorded tokens under the supplied training context. They do not tell us whether a full generated answer will use a tool, cite an eligible source or make a good business decision.

Example to use: If the second target probability rose from 0.5 to 0.8, both losses would be about 0.223 and the mean would also be 0.223.

Assumptions and limits: The probabilities are illustrative, natural logarithms are used, both masks equal one and values are rounded. No model experiment is being reported.

Ask the room: What happens to the denominator if we add another scored target position?

Slide 17 · Check 1: what does a zero loss mask mean?

Main point: Check the distinction between loss and attention.

Give the room a moment to vote before showing the answer. Ask someone to explain why their choice follows from the previous calculation. A common confusion is that zero mask means the prompt disappears. Another is that the mask is a probability predicted by the model. Neither is correct here. The mask is supplied by the training pipeline and selects which errors contribute directly to the objective.

Example to use: Return to the day-extraction record: the instruction remains useful even when only the response contributes direct loss.

Assumptions and limits: The question refers to the response-only loss mask defined in these slides, not an attention mask or a special implementation with different conventions.

Ask the room: Ask for A, B or C and one sentence of reasoning.

Slide 18 · Answer: B - excluded from direct loss

Main point: Resolve the misconception without adding new notation.

B is correct. The model still reads the prompt and uses it to predict the answer. We simply do not add a direct prediction-loss term for those prompt positions. Think of a worked exercise: the question is visible while you are marked on the answer. That analogy is not a description of all neural computations, but it captures visibility versus scoring. Training can still modify computations that use the prompt because those computations affect the scored answer.

Example to use: Changing Extract the day to Extract the location should change the desired response even if both instructions have zero direct loss.

Assumptions and limits: A zero mask removes that position from this loss reduction. It does not guarantee the position has no effect on gradients elsewhere.

Ask the room: Ask someone to contrast this with an attention mask in plain language.

Slide 19 · Why not update every parameter?

Main point: Motivate LoRA through resource and maintenance costs.

Full fine-tuning allows the large original set of weights to move. Training must then handle their gradients and optimizer state, not just store the weights. It can also leave a large model copy to manage for every specialized task. LoRA addresses this engineering burden by keeping the base fixed and learning a small added correction. The motivation is feasibility: can we make useful task changes with fewer trainable values? It is not that changing all parameters is intrinsically wrong.

Example to use: A firm might maintain several small task adapters over one common base instead of storing several independently fine-tuned full checkpoints.

Assumptions and limits: Actual memory and cost depend on optimizer, precision, activation storage, batching and implementation. Fewer trainable parameters do not automatically imply a proportional wall-clock speedup.

Ask the room: Which resources would still be needed even if the original weights stay frozen?

Slide 20 · LoRA: low-rank adaptation

Main point: Read the original LoRA figure rather than infer from a slogan.

The blue block is the existing weight transformation. It remains fixed. The orange blocks are A and B, which form the trainable correction. Follow x at the bottom: it travels through both paths. A maps it into a smaller intermediate representation; B expands it back to the output dimension. At the plus sign the correction is added to the original output h. This happens inside a selected linear layer. It is not an extra language model between layers.

Example to use: Imagine adjusting a large established calculation with a small configurable correction, while keeping the original calculation intact. That is the role of the added branch, not a literal rule-based correction.

Assumptions and limits: The original figure uses a conventional initialization in which B starts at zero, so the added effect initially vanishes. A and B then learn through backpropagation. Frozen W can still be used in differentiable computation.

Ask the room: Point to the figure and ask which blocks receive parameter updates. Source: Hu et al., LoRA, Figure 1.

Slide 21 · LoRA equation: base transformation + correction

Main point: Connect each matrix to the two paths in the figure.

W0 is the original matrix and x is its input. The first term is the original output. In the second term, A reduces the input to r coordinates and B maps those coordinates back to the output size. Alpha divided by r is the scale in the conventional parameterization shown. The product BA has rank at most r, so the possible correction is restricted. The optimizer learns the entries of A and B from the chosen loss. It does not select a few original W entries to unlock.

Example to use: For an input and output dimension of 4096, a rank-eight adapter has 8 times 4096 values in A and the same in B: 65,536 added trainable values.

Assumptions and limits: Target matrices are fixed, W0 is frozen, dimensions are compatible and the update is restricted to BA with rank at most r. This does not prove that the task-optimal update is low rank. Biases or other trainable modules are excluded from this count.

Ask the room: What changes if r doubles while the target matrix dimensions remain fixed?

Slide 22 · What changes when rank increases?

Main point: Explain capacity and cost without promising improvement.

Rank controls the size of the intermediate adapter representation. With a fixed pair of input and output dimensions, doubling rank doubles the number of entries in A and B. That permits a richer correction but increases trainable storage and related training work. It does not guarantee better generalization. Too little rank may restrict the change we need; more rank may be unnecessary or fit peculiarities of a small dataset. Compare choices on separate validation cases and protect a final test set.

Example to use: Rank four adds 32,768 entries to one 4096 by 4096 transformation. Rank eight adds 65,536. These are counts for one target matrix, not the whole transformer.

Assumptions and limits: The count assumes the standard two-factor adapter with no additional trainable biases. A higher permitted rank is a larger update class, not a claim that the learned product attains that rank.

Ask the room: Would a larger rank be justified by a lower training loss alone?

Slide 23 · Who chooses where LoRA is applied?

Main point: Separate design choices from values learned by gradients.

The developer chooses which named modules receive adapters. Common targets include attention query and value projections, but keys, outputs and feed-forward transformations are also possible. The rank and scaling are configuration choices. After those choices are fixed, backpropagation computes gradients and the optimizer learns the numerical entries in A and B. The software does not automatically discover a scientifically optimal list of layers from the task name.

Example to use: The example configuration targets q_proj and v_proj at rank eight. Adding k_proj would be another experimental condition to evaluate, not a conclusion that keys are always necessary.

Assumptions and limits: Module names and shapes depend on the model architecture. Target selection should be compared under matched data, training budgets and held-out tasks. No one-to-one mapping from a business task to Q, K or V is assumed.

Ask the room: What evidence would justify changing the target-module list?

Slide 24 · Why QLoRA? LoRA still keeps a large base in memory

Main point: Explain why adapters do not eliminate base-weight memory.

LoRA reduces the parameters we train, but the base still has to be available for the forward computation. A large frozen base can therefore remain the dominant memory cost. The bar chart isolates raw weight storage. Seven billion weights at sixteen bits require fourteen billion bytes, or fourteen decimal gigabytes. Four bits per weight gives three and a half gigabytes before overhead. Quantization attacks this remaining storage problem. It may allow the model to fit on a smaller GPU or leave room for longer context and a larger batch.

Example to use: A base that nearly fills a sixteen-gigabyte GPU at higher precision leaves little room for activations. A compact frozen representation can make adaptation feasible where plain LoRA would run out of memory.

Assumptions and limits: This illustration uses seven billion weights and decimal GB. It excludes quantization scales, metadata, adapters, gradients, optimizer state, activations and runtime allocations. Actual total memory is higher; cost savings are workload-dependent.

Ask the room: Why is the three-and-a-half-gigabyte number not a complete training memory budget?

Slide 25 · What does storing a weight in 4 bits mean? - 1 of 3

Main point: A large frozen base still occupies memory

LoRA reduces what is trained, but the original base weights remain needed for computation. Four bits can encode sixteen distinct codes. In quantized storage, a code identifies an approximate numerical level, with scaling information for blocks of weights. The original continuous-looking range is represented more compactly. This is a storage approximation, not a statement that the original model had only sixteen different weights. During computation the implementation can reconstruct or use higher-precision values. The adapters can also be stored and trained at a different precision.

Example to use: Think of storing a rounded measurement plus a scale rather than its full original numerical representation. The smaller record saves space but introduces approximation. QLoRA is designed to train adapters while using that compact frozen base.

Assumptions and limits: The exact quantizer matters. QLoRA commonly uses NF4 with block scaling, not a universal equally spaced list of sixteen values. Four-bit base storage does not imply four-bit gradients or arithmetic everywhere.

Ask the room: What do we give up when we compress numerical weights?

Slide 26 · What does storing a weight in 4 bits mean? - 2 of 3

Main point: Four bits encode sixteen codes

Codes select approximate numerical levels, with scaling information. This compact representation saves base-weight storage. Four bits can encode sixteen distinct codes. In quantized storage, a code identifies an approximate numerical level, with scaling information for blocks of weights. The original continuous-looking range is represented more compactly. This is a storage approximation, not a statement that the original model had only sixteen different weights. During computation the implementation can reconstruct or use higher-precision values. The adapters can also be stored and trained at a different precision.

Example to use: Think of storing a rounded measurement plus a scale rather than its full original numerical representation. The smaller record saves space but introduces approximation. QLoRA is designed to train adapters while using that compact frozen base.

Assumptions and limits: The exact quantizer matters. QLoRA commonly uses NF4 with block scaling, not a universal equally spaced list of sixteen values. Four-bit base storage does not imply four-bit gradients or arithmetic everywhere.

Ask the room: What do we give up when we compress numerical weights?

Slide 27 · What does storing a weight in 4 bits mean? - 3 of 3

Main point: Storage precision is not all computation

Adapters and calculations can use higher precision. Quantization adds approximation and does not guarantee the same quality or speed. Four bits can encode sixteen distinct codes. In quantized storage, a code identifies an approximate numerical level, with scaling information for blocks of weights. The original continuous-looking range is represented more compactly. This is a storage approximation, not a statement that the original model had only sixteen different weights. During computation the implementation can reconstruct or use higher-precision values. The adapters can also be stored and trained at a different precision.

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Ask the room: What do we give up when we compress numerical weights?

Slide 28 · QLoRA: LoRA with a quantized base

Main point: Read the original memory diagram and identify both savings.

The original figure compares three configurations. On the left, full fine-tuning has a large optimizer-state burden because many weights are trained. In the middle, LoRA trains only adapters, reducing that burden, but keeps a sixteen-bit base. On the right, QLoRA also reduces base-weight storage to four bits. The green arrows show gradients flowing through computation; the blue arrows show which parameters receive updates. Those are different ideas. The paged optimizer addresses memory spikes by moving suitable optimizer data between memory spaces.

Example to use: The practical question is not which acronym is newer. It is whether the desired model, context length and training batch fit the available hardware at an acceptable cost.

Assumptions and limits: The original paper diagram is a conceptual memory comparison, not a to-scale total-memory measurement. Numerical quality, speed and paging overhead depend on the implementation and task. A frozen quantized base remains part of the model computation.

Ask the room: Ask which component changes between ordinary LoRA and QLoRA. Source: Dettmers et al., QLoRA, Figure 1.

Slide 29 · Check 2: can LoRA be used with SFT and DPO?

Main point: Check whether objective and parameterization remain separate.

Ask for a vote before revealing the answer. Participants should use the three-question map from the start: what is rewarded, which parameters change and how the base is stored. Nothing in the low-rank matrix formula requires that the loss come from only one demonstrated answer. The same kind of adapter can receive gradients from a preference loss. This question is a useful checkpoint before introducing the extra moving parts in preference training.

Example to use: An SFT record can update A and B toward a demonstrated answer. A DPO pair can update A and B toward a relative preference.

Assumptions and limits: We assume the model, trainer and PEFT implementation support the selected combination. Conceptual compatibility is not a guarantee of a good result.

Ask the room: Ask for a one-sentence explanation, not just the letter C.

Slide 30 · Answer: C - objective and update method combine

Main point: State the distinction in language that can be repeated.

C is correct. SFT or DPO supplies the loss. LoRA specifies which added matrices are trainable. The optimizer updates those matrices using gradients of that loss. QLoRA additionally changes how the frozen base is represented in memory. A useful way to say this is: the objective says what to encourage, while the adapter says where the adjustable capacity lives. We now return from the engineering question to the learning-signal question.

Example to use: One experiment can use SFT plus LoRA. A second can start from an appropriate checkpoint and use DPO plus LoRA, but it needs preference pairs rather than only demonstrations.

Assumptions and limits: The table is a map of compatible choices. It does not assert that the same data or hyperparameters should be used for all objectives.

Ask the room: Which additional fields must a DPO dataset contain?

Slide 31 · Why move beyond demonstrations?

Main point: Motivate preferences by the burden of writing ideal outputs.

For some tasks it is hard to write a single perfect answer but much easier to say which of two answers is better. Experts may be able to verify an explanation faster than they can produce one from scratch. This makes comparison data useful. The comparison must still be about a defined criterion: factual support, completeness, clarity or another objective. Without that criterion, labels can reward style and confidence rather than substance.

Example to use: A reviewer compares two summaries. One is slightly shorter and accurately reflects the source; the other is polished but adds an unsupported claim. Choosing the first records useful feedback without asking the reviewer to rewrite it.

Assumptions and limits: Comparisons are not automatically cheaper, more accurate or less biased. The collection design and reviewer expertise matter. A preference label does not certify that either answer is wholly correct.

Ask the room: In your work, which outputs are easier to judge than to write?

Slide 32 · RLHF: the goal; PPO and DPO: different routes

Main point: Resolve the terminology before comparing algorithms.

RLHF means reinforcement learning from human feedback, but it is often used loosely for a broader family of preference-based alignment methods. Here we compare two specific recipes. The conventional route first learns a reward predictor from comparisons, then uses an RL procedure such as PPO to improve a response-generating model. Standard DPO instead updates the response model directly using recorded preference pairs. It avoids the separate reward-model training stage and online generation loop in that recipe. DPO is therefore not the opposite of listening to human feedback.

Example to use: Both routes can start from the same collection of preferred and rejected summaries. The difference is what is trained and how the feedback reaches the language model.

Assumptions and limits: This comparison refers to conventional reward-model-plus-PPO RLHF and standard offline DPO. Other variants exist and may add generation, different losses or other feedback sources.

Ask the room: Which route has a separately trained reward model in the version shown?

Slide 33 · RLHF: reinforcement learning from human feedback

Main point: Explain the reward-model route using the original illustration.

Start at the preference data on the left. Reviewers compare answers. Those labels train a reward model: a predictor of which responses reviewers favor. The language model then generates responses, receives scores from that reward predictor and is updated to obtain better scores. PPO is one way to perform this policy optimization. The model changes as the process proceeds, so generated responses and scores must be refreshed. A reference constraint discourages excessive departures from the starting behavior.

Example to use: A summary that receives a higher learned score is more strongly encouraged. If the reward predictor mistakenly treats length as quality, the language model may learn to produce longer summaries without becoming more accurate.

Assumptions and limits: The reward is a proxy for human preference, not observed truth or business value. The optimization procedure and reference penalty must be specified. Source illustration: Rafailov et al., DPO, Figure 1.

Ask the room: What could go wrong if the reward model has a systematic shortcut?

Slide 34 · DPO removes the reward-model training loop

Main point: Explain the practical simplification before the equation.

The original DPO panel has a shorter path. Recorded preference pairs are fed directly into a training loss for the language model. There is no separately trained reward predictor in standard DPO, and it does not need to repeatedly generate fresh responses during the training update loop. That can reduce engineering complexity and training overhead. The loss uses the model and a fixed reference to express a preference comparison. We will unpack that reference next rather than assume it is obvious.

Example to use: Take a fixed dataset of chosen and rejected summaries. DPO can train on those rows directly. The conventional route would first use them to learn a reward model and then optimize generated outputs against its scores.

Assumptions and limits: This is standard offline DPO. Responses still need to be generated or collected when building the dataset, and some variants refresh data. Simpler machinery does not guarantee better task performance.

Ask the room: Ask someone to name the two components removed from the conventional training recipe.

Slide 35 · The original lecture makes the same comparison

Main point: Consolidate the two routes using the preserved lecture material.

Read the source slide from top to bottom. Its central question is whether preference training can avoid the reward-model and on-policy machinery of the PPO route. The bottom comparison puts the two paths next to one another. Use it to repeat the operational distinction, not to introduce another method. The phrase positive gradient on good and negative gradient on bad is shorthand: the actual DPO loss weights a relative preference margin. It is not two independent instructions to push probabilities by arbitrary amounts.

Example to use: For one prompt, compare a supported response with an unsupported response. DPO increases the preferred-versus-rejected likelihood ratio relative to a reference model.

Assumptions and limits: The source lecture summarizes standard DPO. Its informal gradient description is unpacked by the following equation and should not be treated as a complete specification. Source: Stanford CS336 Lecture 15, PDF page 55, which reproduces the DPO comparison figure.

Ask the room: Which path would you choose if your main constraint were operating a simpler training pipeline? What would you still need to test?

Slide 36 · A DPO record: the preference label is explicit

Main point: Show what the dataset rows actually contain.

Each row has one prompt, one chosen response and one rejected response. The label is encoded by which response occupies which field. In the first row, the evidence says the meeting moved to Friday, so Friday is preferred to Tuesday. In the second, no day is supplied, so acknowledging the missing information is preferred to inventing Tuesday. Both rows need the full same context for both candidates. A response pair from different questions would not express the intended comparison.

Example to use: A JSON record can contain prompt: No meeting day is provided. Which day?; chosen: The day is not specified.; rejected: Tuesday. No separate numeric truth-probability label is implied.

Assumptions and limits: This illustration uses strict binary preferences and no ties. Labels reflect a defined review criterion. Contradictory labels or pairs with two unacceptable answers need a deliberate handling policy.

Ask the room: Could two reviewers disagree because one rewards brevity and the other rewards explanation?

Slide 37 · The reference model is a frozen starting point

Main point: Explain the reference as a model copy with a specific role.

We start with a model checkpoint, commonly one that already follows instructions after SFT. Conceptually make two copies. One stays fixed and supplies reference probabilities for the candidate responses. The other is updated during preference training. Both evaluate the same prompt and responses. The reference helps express how the new preference ratio differs from the starting model. It is not a reviewer, a truth database or the dataset itself.

Example to use: If the starting model prefers a rejected answer two to one, we can measure how training changes that ratio. DPO compares the change relative to that fixed starting preference.

Assumptions and limits: The reference is fixed in the standard objective shown and must assign nonzero likelihood to scored candidates. LoRA may separately freeze base matrices while training adapters; that is a different use of the word frozen. Implementations can share storage without changing the conceptual roles.

Ask the room: Ask which copy changes after a gradient update.

Slide 38 · The reference model is a frozen starting point - 1 of 3

Main point: Begin with a useful checkpoint

Often this is an already instruction-following model. A new local SFT stage is common but not always required. We start with a model checkpoint, commonly one that already follows instructions after SFT. Conceptually make two copies. One stays fixed and supplies reference probabilities for the candidate responses. The other is updated during preference training. Both evaluate the same prompt and responses. The reference helps express how the new preference ratio differs from the starting model. It is not a reviewer, a truth database or the dataset itself.

Example to use: If the starting model prefers a rejected answer two to one, we can measure how training changes that ratio. DPO compares the change relative to that fixed starting preference.

Assumptions and limits: The reference is fixed in the standard objective shown and must assign nonzero likelihood to scored candidates. LoRA may separately freeze base matrices while training adapters; that is a different use of the word frozen. Implementations can share storage without changing the conceptual roles.

Ask the room: Ask which copy changes after a gradient update.

Slide 39 · The reference model is a frozen starting point - 2 of 3

Main point: Keep one reference fixed

It scores the same candidate answers without updating. It is not the reviewer, dataset or source of factual truth. We start with a model checkpoint, commonly one that already follows instructions after SFT. Conceptually make two copies. One stays fixed and supplies reference probabilities for the candidate responses. The other is updated during preference training. Both evaluate the same prompt and responses. The reference helps express how the new preference ratio differs from the starting model. It is not a reviewer, a truth database or the dataset itself.

Example to use: If the starting model prefers a rejected answer two to one, we can measure how training changes that ratio. DPO compares the change relative to that fixed starting preference.

Assumptions and limits: The reference is fixed in the standard objective shown and must assign nonzero likelihood to scored candidates. LoRA may separately freeze base matrices while training adapters; that is a different use of the word frozen. Implementations can share storage without changing the conceptual roles.

Ask the room: Ask which copy changes after a gradient update.

Slide 40 · The reference model is a frozen starting point - 3 of 3

Main point: Update the response model

DPO measures how the chosen-versus-rejected likelihood ratio changes relative to that reference. We start with a model checkpoint, commonly one that already follows instructions after SFT. Conceptually make two copies. One stays fixed and supplies reference probabilities for the candidate responses. The other is updated during preference training. Both evaluate the same prompt and responses. The reference helps express how the new preference ratio differs from the starting model. It is not a reviewer, a truth database or the dataset itself.

Example to use: If the starting model prefers a rejected answer two to one, we can measure how training changes that ratio. DPO compares the change relative to that fixed starting preference.

Assumptions and limits: The reference is fixed in the standard objective shown and must assign nonzero likelihood to scored candidates. LoRA may separately freeze base matrices while training adapters; that is a different use of the word frozen. Implementations can share storage without changing the conceptual roles.

Ask the room: Ask which copy changes after a gradient update.

Slide 41 · DPO rewards a relative preference shift - 1 of 3

Main point: Compare two answers to the same prompt

The dataset identifies a chosen and a rejected response. A preference criterion explains why one is favored. X is the shared prompt and evidence. Y plus is the chosen response and y minus the rejected response. The first log ratio describes how strongly the current model favors chosen over rejected. The second is the same ratio under the frozen reference. Delta is their difference. The sigmoid and beta turn that margin into a pairwise training loss. Reducing the loss encourages a larger relative preference for the chosen response. Candidate response likelihoods are sequence scores, built from conditional token probabilities.

Example to use: If the reference chosen-to-rejected ratio is one half and the current ratio is two, the ratio has improved fourfold. Delta is the natural logarithm of four. These are illustrative ratios, not measured confidence in correctness.

Assumptions and limits: The standard objective uses strict preference pairs, a fixed reference, compatible sequence scoring and positive reference likelihood on candidates. Beta is positive. A scalar-reward logistic preference model motivates the formulation; it is not guaranteed to match all human judgments.

Ask the room: Does the sigmoid output prove that the chosen answer is factually correct?

Slide 42 · DPO rewards a relative preference shift - 2 of 3

Main point: Compare current and starting preferences

Both models score both responses. The loss rewards improving the chosen/rejected ratio relative to the fixed reference. X is the shared prompt and evidence. Y plus is the chosen response and y minus the rejected response. The first log ratio describes how strongly the current model favors chosen over rejected. The second is the same ratio under the frozen reference. Delta is their difference. The sigmoid and beta turn that margin into a pairwise training loss. Reducing the loss encourages a larger relative preference for the chosen response. Candidate response likelihoods are sequence scores, built from conditional token probabilities.

Example to use: If the reference chosen-to-rejected ratio is one half and the current ratio is two, the ratio has improved fourfold. Delta is the natural logarithm of four. These are illustrative ratios, not measured confidence in correctness.

Assumptions and limits: The standard objective uses strict preference pairs, a fixed reference, compatible sequence scoring and positive reference likelihood on candidates. Beta is positive. A scalar-reward logistic preference model motivates the formulation; it is not guaranteed to match all human judgments.

Ask the room: Does the sigmoid output prove that the chosen answer is factually correct?

Slide 43 · DPO rewards a relative preference shift - 3 of 3

Main point: Do not interpret preference as truth

A lower DPO loss means better fit to preference labels. It does not establish factual accuracy, calibrated beliefs or better business outcomes. X is the shared prompt and evidence. Y plus is the chosen response and y minus the rejected response. The first log ratio describes how strongly the current model favors chosen over rejected. The second is the same ratio under the frozen reference. Delta is their difference. The sigmoid and beta turn that margin into a pairwise training loss. Reducing the loss encourages a larger relative preference for the chosen response. Candidate response likelihoods are sequence scores, built from conditional token probabilities.

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Ask the room: Does the sigmoid output prove that the chosen answer is factually correct?

Slide 44 · DPO: connect the explanation to the equation

Main point: Read the preference margin without turning it into truth confidence.

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Ask the room: Does the sigmoid output prove that the chosen answer is factually correct?

Slide 45 · What assumptions support the DPO equation?

Main point: Separate a useful statistical model from a universal description of preferences.

The standard derivation imagines a scalar reward for a response in its context. A logistic pairwise model relates reward differences to preference probabilities. Combining that preference model with a KL-regularized reward objective gives the DPO form. These are modeling assumptions that make the relationship tractable. Reviewers can have inconsistent priorities, preferences can depend on omitted context and pairs can contain ties. The training computation can still fit such labels, but the idealized interpretation becomes less reliable.

Example to use: One reviewer prefers a short answer and another prefers a detailed answer. Treating the labels as one consistent scalar ranking may hide that difference unless the task and rubric specify the intended trade-off.

Assumptions and limits: The binary form omits ties, the reference is fixed and candidate likelihoods are well-defined. The preference population and collection process should be documented. Neither the derivation nor a low loss certifies truth or operational safety.

Ask the room: What would you record when two qualified reviewers systematically disagree?

Slide 46 · Why use DPO after SFT? - 1 of 3

Main point: SFT establishes a response pattern

Reviewed demonstrations can teach instruction following, output structure and a specific interpretation skill. SFT can establish a useful starting response pattern: following instructions, using a format or carrying out a task. Preference training can then adjust recurring differences between plausible responses. It is common to begin DPO from an SFT-tuned or already instruction-tuned checkpoint because that gives a reasonable starting distribution. But a new local SFT stage is not logically required in every project. If demonstrations already produce the desired behavior, adding DPO may be unnecessary.

Example to use: First teach concise source-based summaries using reviewed demonstrations. Then use preference pairs to discourage recurring unsupported embellishments, provided those pairs add useful information beyond the demonstrations.

Assumptions and limits: The benefit depends on the starting checkpoint, preference-data coverage and task. SFT can also teach preference-related behavior; the distinction is the data and objective, not a rigid division of capabilities.

Ask the room: What observed failure would justify the extra preference-training stage?

Slide 47 · Why use DPO after SFT? - 2 of 3

Main point: DPO adjusts recurring preferences

Comparisons can favor supported, useful responses over plausible alternatives without requiring a new ideal answer every time. SFT can establish a useful starting response pattern: following instructions, using a format or carrying out a task. Preference training can then adjust recurring differences between plausible responses. It is common to begin DPO from an SFT-tuned or already instruction-tuned checkpoint because that gives a reasonable starting distribution. But a new local SFT stage is not logically required in every project. If demonstrations already produce the desired behavior, adding DPO may be unnecessary.

Example to use: First teach concise source-based summaries using reviewed demonstrations. Then use preference pairs to discourage recurring unsupported embellishments, provided those pairs add useful information beyond the demonstrations.

Assumptions and limits: The benefit depends on the starting checkpoint, preference-data coverage and task. SFT can also teach preference-related behavior; the distinction is the data and objective, not a rigid division of capabilities.

Ask the room: What observed failure would justify the extra preference-training stage?

Slide 48 · Why use DPO after SFT? - 3 of 3

Main point: Do not add a stage without a need

SFT may already be sufficient. Add preference training only when reliable labels address a measurable recurring failure. SFT can establish a useful starting response pattern: following instructions, using a format or carrying out a task. Preference training can then adjust recurring differences between plausible responses. It is common to begin DPO from an SFT-tuned or already instruction-tuned checkpoint because that gives a reasonable starting distribution. But a new local SFT stage is not logically required in every project. If demonstrations already produce the desired behavior, adding DPO may be unnecessary.

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Ask the room: What observed failure would justify the extra preference-training stage?

Slide 49 · SFT, DPO and conventional RLHF: what differs?

Main point: Leave participants with a usable comparison of learning signals.

SFT imitates reviewed response tokens. DPO uses chosen and rejected response pairs with a fixed reference in its loss. Conventional RLHF first trains a reward predictor from feedback and then updates a policy using scores on generated responses. The methods overlap in purpose but require different data and machinery. LoRA can be used to reduce trainable parameters in suitable implementations; it does not decide which of these signals is appropriate.

Example to use: If we have ideal answers, SFT is the natural starting point. If we have reliable comparisons, DPO is a practical route. If we need a reward-based generation-and-optimization process, the conventional RL recipe may be relevant.

Assumptions and limits: This table describes standard versions, not every research variant. Compare performance on the actual intended task rather than assume a universal ranking of methods.

Ask the room: Ask each participant which data they already possess: demonstrations, comparisons or outcome-based feedback.

Slide 50 · Do training records have to be i.i.d.?

Main point: Distinguish computing a loss from justifying statistical claims.

Independent and identically distributed means records do not depend on one another and come from the same distribution. LoRA is only a way to represent an update, so it has no intrinsic record-independence assumption. We can also compute SFT or DPO losses on correlated records. The problem appears when we interpret sample size or claim generalization. Repeated examples from one source may provide far less independent information than their row count suggests. Related records must stay together in data splits.

Example to use: One source document is paraphrased one hundred times. Randomly putting some paraphrases in training and others in test can make the model look better on unseen material than it really is.

Assumptions and limits: Simple independent-sample generalization or uncertainty arguments need an appropriate independence or dependence-control assumption. Token dependencies are already modeled autoregressively and are a separate issue. Shuffling rows does not make correlated source families independent.

Ask the room: What is the independent unit in your dataset: a row, document, company, customer or time period?

Slide 51 · Preference improvement is not probability calibration - 1 of 3

Main point: Preferred means favored under a rubric

A reviewer may prefer clarity or brevity. That preference is not automatically a check of every factual statement. A preferred answer is one a reviewer favors under a rubric. A correct answer agrees with the relevant facts. A calibrated probability matches observed frequencies for a defined event. These are separate properties. Training can make a model sound more careful or confident without making its numerical confidence valid. To assess a probability we need a clearly defined event, resolved labels and appropriate independent evaluation, not merely votes on how the answer reads.

Example to use: Among one hundred cases assigned about eighty percent probability of a correct extraction, we would expect roughly eighty correct extractions under good calibration. A preference for confident phrasing does not establish that relationship.

Assumptions and limits: Calibration is distribution- and event-specific. It requires suitable labels and sufficient relevant cases; dependencies and drift affect estimates. No automatic calibration guarantee follows from SFT, DPO or LoRA.

Ask the room: What outcome label would you need before accepting an eighty-percent claim?

Slide 52 · Preference improvement is not probability calibration - 2 of 3

Main point: Correct means supported and accurate

A clear answer can still be false. A correct categorical answer does not validate its confidence number. A preferred answer is one a reviewer favors under a rubric. A correct answer agrees with the relevant facts. A calibrated probability matches observed frequencies for a defined event. These are separate properties. Training can make a model sound more careful or confident without making its numerical confidence valid. To assess a probability we need a clearly defined event, resolved labels and appropriate independent evaluation, not merely votes on how the answer reads.

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Ask the room: What outcome label would you need before accepting an eighty-percent claim?

Slide 53 · Preference improvement is not probability calibration - 3 of 3

Main point: Calibrated means matching frequencies

For a defined event, cases scored near 80 percent should resolve positively about 80 percent of the time in the relevant population. A preferred answer is one a reviewer favors under a rubric. A correct answer agrees with the relevant facts. A calibrated probability matches observed frequencies for a defined event. These are separate properties. Training can make a model sound more careful or confident without making its numerical confidence valid. To assess a probability we need a clearly defined event, resolved labels and appropriate independent evaluation, not merely votes on how the answer reads.

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Assumptions and limits: Calibration is distribution- and event-specific. It requires suitable labels and sufficient relevant cases; dependencies and drift affect estimates. No automatic calibration guarantee follows from SFT, DPO or LoRA.

Ask the room: What outcome label would you need before accepting an eighty-percent claim?

Slide 54 · Check 3: what would justify using the update?

Main point: Check whether training evidence is being confused with deployment evidence.

Let participants choose before revealing the answer. Lower training loss tells us the model fits the training targets better. Reviewer preference can tell us the responses meet a stated rubric better. Neither alone establishes transfer to new workflow cases or preservation of required tool behavior. The deployment-oriented choice is held-out task improvement with the necessary checks still working. Ask for the observation that would make each weaker answer useful but insufficient.

Example to use: A model can write better explanations yet stop calling the calculator. That would be visible only if tool execution is evaluated separately from prose.

Assumptions and limits: The comparison must use the same cases, context, tools and material execution settings, with no tuning on the final test.

Ask the room: Which two failure categories would you inspect before releasing an updated model?

Slide 55 · Answer: C - test behavior in the workflow

Main point: Look for which stage improved or deteriorated.

C is the best answer. Evaluate the model in the context where it will be used. Check whether it extracts the right fact, requests the right tool with the right inputs and respects the returned result. A final-answer score can hide opposite changes in these stages. Keep base and adapted conditions matched so the difference can reasonably be attributed to the update rather than to different source access or tools.

Example to use: An adapter improves date extraction but generates a final summary that contradicts the extracted date. The right repair is not established by the average score; the trace reveals where the inconsistency occurs.

Assumptions and limits: Workflow checks need reviewed expected behavior. Held-out cases must not have guided training or model selection. This slide proposes an evaluation design rather than reporting new measured gains.

Ask the room: Ask which checkpoint would catch a correct calculator result followed by a wrong written answer.

Slide 56 · Class exercise: identify what each setting changes

Main point: Use three small records to verify conceptual understanding.

This is an inspection exercise rather than a live GPU training session. Start with the SFT record and mark which response positions are scored. Next compare rank four with rank eight for one fixed target matrix. Finally read the chosen and rejected responses and state exactly what the preference reviewer rewards. Participants can work from the slides. The optional formula notebook is a companion calculation, not evidence that a new training run has occurred.

Example to use: For the missing-day preference row, a good explanation is: the label rewards acknowledging missing information rather than inventing a day. It is more specific than saying the chosen answer is better.

Assumptions and limits: The exercise illustrates declared configurations. The optional notebook may use a different concrete record, so compare concepts rather than assume identical inputs or measured results.

Ask the room: Ask a pair to explain one answer in everyday language without using an acronym.

Slide 57 · Exercise review: three distinct answers

Main point: Resolve the three recurring misunderstandings.

First, response-only loss still lets the model read its prompt. Second, rank controls the size of the added factor matrices; it does not unlock that many original weights. Third, a preference label must encode a stated reason for favoring one response over another. Each answer corresponds to a different decision in a training project. If these distinctions are clear, participants can read a basic training configuration without confusing data, objective and parameterization.

Example to use: Rank eight has twice as many adapter entries as rank four for the same target dimensions. That numerical fact says nothing by itself about which output a reviewer should prefer.

Assumptions and limits: The parameter count excludes optional trainable biases and other modules. The mask convention is the response-only convention already declared.

Ask the room: Which answer would change if we chose equal-example rather than equal-token loss weighting?

Slide 58 · Choose the intervention that matches the problem

Main point: Avoid prescribing training for every failure.

If information is absent, investigate source access and retrieval. If valid context is repeatedly misinterpreted, inspect instructions and demonstrations and consider SFT. If outputs violate a clear preference, reviewed comparisons may motivate DPO. If an action exceeds authority, enforce permissions in the application or infrastructure. Training can change tendencies, but it does not create missing evidence or provide an enforceable authority boundary.

Example to use: A system that releases a document after a grant was revoked needs a current permission check. A new preference dataset saying do not release is not an equivalent control.

Assumptions and limits: These are diagnostic starting points, not exclusive remedies. Real failures can span several components and require controlled retesting after changes.

Ask the room: Name a failure for which changing weights would be the wrong first move.

Slide 59 · Objectives reviewed - next, run the experiment

Main point: Connect the explanations to the practical adaptation session.

We can now say what an SFT record contains, why response masks appear in the loss, what LoRA changes and why quantization may be needed. We can distinguish a frozen reference from a frozen LoRA base and explain how direct preference training differs from the conventional reward-model route. The next step is to make those choices visible in a real or recorded run and compare the resulting behavior on unseen cases. The goal is not to demonstrate that every update helps.

Example to use: In the lab, inspect one actual training record, the adapter configuration and one held-out failure before drawing conclusions from a learning curve.

Assumptions and limits: Any result must be tied to its dataset, model revision, objective, configuration and evaluation split. The conceptual examples in this lecture are not a claim of improved financial performance.

Ask the room: Ask a participant to explain the distinction between what is rewarded and which parameters change in one sentence.